Project management formulas and metrics are **quantitative tools** used to measure, monitor, and control the performance of a project. They provide managers with objective data to make informed decisions about cost, schedule, risk, and resource efficiency.

Together, these formulas and metrics form the **analytical /quantitative backbone of project management**, enabling managers to:

- Measure **progress** (Earned Value, SPI).
- Forecast **costs** (ETC, EAC).
- Assess **risks** (Probability × Impact).
- Evaluate **value** (ROI).
- Optimize **resources** (Utilization).

📊 Earned Value Formulas

- **Definition:** A set of performance metrics that measure project progress in terms of cost and schedule.
- **Key Formulas:**
  - **EV (Earned Value):** % Complete × Budget. (Value of work actually completed)
  - **CPI (Cost Performance Index):** EV ÷ AC (Actual Cost/ cost efficiency).
  - **SPI (Schedule Performance Index):** EV ÷ PV (Planned Value/ schedule efficiency).
- **Purpose:** Quantifies how efficiently resources are being used and whether the project is on track. (Shows if the project is on budget and on schedule.)

🏗️ Critical Path

- **Definition:** The longest chain of dependent tasks that determines the shortest possible project duration.
- **Formula:** Float = LFT (Latest Finish Time) – EST (Earliest Start Time).
- **Purpose:** Identifies tasks that cannot be delayed without delaying the entire project.

💰 Estimate to Complete (ETC)

- **Definition:** Forecast of the cost required to finish the remaining project work.
- **Formula:** ETC = (BAC – EV) ÷ CPI.
- **Purpose:** Helps predict future spending and manage budgets.

⚠️ Risk Probability & Impact

- **Definition:** A method to assess risks by multiplying their likelihood by their potential impact.
- **Formula:** Risk Score = Probability × Impact.
- **Purpose:** Prioritizes risks so teams focus on the most critical ones.

📈 Project ROI

- **Definition:** Return on Investment measures the profitability of a project.
- **Formula:** ROI (%) = (Net Benefit ÷ Total Cost) × 100.
- **Purpose:** Evaluates whether the project delivers sufficient financial value compared to its cost.

👥 Resource Utilization

- **Definition:** Measures how effectively resources (people, equipment, time) are being used.
- **Formula:** Utilization = Actual Hours ÷ Available Hours.
- **Purpose:** Ensures resources are neither underused nor overburdened.

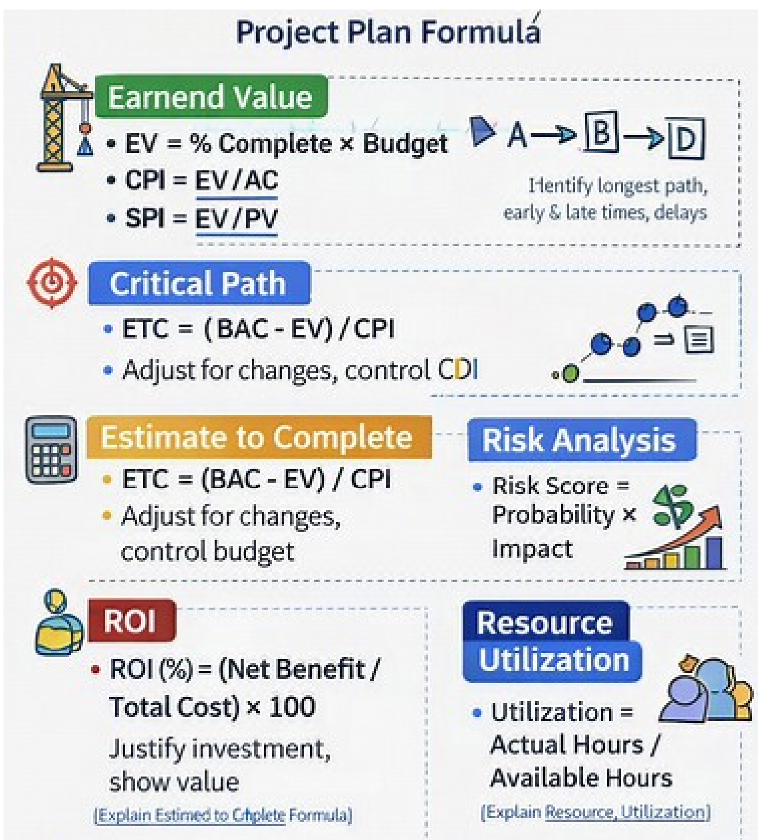
📊 Additional Metrics

- **Definition:** Supplementary measures that provide deeper insights into project health.
- **Examples:**
  - **SPI (Schedule Performance Index):** EV ÷ PV.
  - **TCPI (To-Complete Performance Index):** (BAC – EV) ÷ (BAC – AC).
  - **Burn Rate:** Speed at which budget is consumed.
  - **EAC (Estimate at Completion):** BAC ÷ CPI.
- **Purpose:** Track efficiency, schedule adherence, and completion forecasts.

📊 Project Management Formula (Mathematical Formula)

Category	Key Formula / Concept	Purpose
Earned Value Formulas	EV = % Complete × Budget CPI = EV / AC SPI = EV / PV	Measure cost & schedule performance
Critical Path	Float = LFT – EST	Identify longest path, avoid delays
Estimate to Complete (ETC)	ETC = (BAC – EV) / CPI	Forecast remaining cost & control budget
Risk Probability & Impact	Risk Score = Probability × Impact	Assess, prioritize, and mitigate risks
Project ROI	ROI = (Net Benefit / Total Cost) × 100	Evaluate profitability & justify investment
Resource Utilization	Utilization = Actual Hours / Available Hours	Monitor workload & optimize efficiency
Additional Metrics	SPI, TCPI, CPI, Burn Rate, EAC	Track performance & completion forecasts

✨— These formulas form the analytical backbone of project management, enabling professionals to **quantify progress, control costs, assess risks, and optimize resources** for predictable success.



✨— Mastering these formulas enables precise control over **cost, schedule, risk, and efficiency**, forming the quantitative foundation of successful project management.

- **Earned Value** → EV = % Complete × Budget, CPI = EV / AC, SPI = EV / PV
- **Critical Path** → Float = LFT – EST, identify longest path
- **Estimate to Complete** → ETC = (BAC – EV) / CPI
- **Risk Analysis** → Risk Score = Probability × Impact
- **ROI** → ROI (%) = (Net Benefit / Total Cost) × 100
- **Resource Utilization** → Utilization = Actual Hours / Available Hours

Quick Reference

Metric	**Definition**	**Formula**
Earned Value	Measures progress in terms of cost & schedule စီမံကိန်း၏ တိုတက်မှုကို ကုန်ကျစရိတ်နှင့် အချိန်ဇယားအပေ အခြေခံ၍ တိုင်းတခြင်	EV = % Complete × Budget
CPI	Cost efficiency စီမံကိန်း၏ ကုန်ကျစရိတ် ထိရေက်မှု	CPI = EV ÷ AC
SPI	Schedule efficiency အချိန်ဇယအပေ အောင်မြင်မှု	SPI = EV ÷ PV
Critical Path	Longest chain of dependent tasks စီမံကိန်းအချိန်ဇယားကို သတ်မှတ်သောအရေးကြီးလမ်းကြောင်း	Float = LFT – EST
ETC	Forecast cost to finish remaining work ကျန်ရှိသေ အလုပ်ကို ပြီးစီးရန် လိုအပ်သေ ကုန်ကျစရိတ် ခန့်မှန်ခြင်	ETC = (BAC – EV) ÷ CPI
Risk Probability & Impact	Risk score based on likelihood × impact အန္တရယ်ကို ဖြစ်နိုင်ခြေ × သက်ရောက်မှု ဖြင့် တိုင်တခြင်	Risk = Probability × Impact
Project ROI	Return on Investment စီမံကိန်း၏ အကျိုးအမြတ် အချို	ROI = (Net Benefit ÷ Total Cost) × 100
Resource Utilization	Efficiency of resource use အရင်းအမြစ် အသုံးပြုမှု ထိရေက်မှု	Utilization = Actual Hours ÷ Available Hours
Additional Metrics	Extra measures (EAC, TCPI, Burn Rate) ထပ်ဆောင် စီမံကိန် တိုင်တချက်မျ	EAC = BAC ÷ CPI

✨— This chart helps learners quickly connect **formulas with practical meaning**. For example:

- **CPI < 1** → Over budget.
- **SPI < 1** → Behind schedule.
- **High Risk Score** → Needs immediate mitigation.
- **ROI > 100%** → Project delivers strong value.